CISS350: Data Structures and Advanced Algorithms
Quiz q10603

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For each of the following questions, evaluate the expression written in either prefix/Polish notation or postfix/Reverse Polish notation, or write ERROR if the expression is invalid. You only need to write down the answer. You need not explain.

Q1. Polish notation: `* + 3 5 - 7 2`
Answer:

```
40
```

Q2. Polish notation: `+ / 8 2 - 4 * 7 2`
Answer:

```
-6
```

Q3. Polish notation: `* 8 * 2 + 3 - 2 / 7 2`
Answer:

```
32
```

Q4. Polish notation: `/ + - * 2 3 7 4 2`
Answer:

```
0
```

Q5. Reverse Polish notation: `2 3 4 * 5 * +`
Answer:

```
62
```

Q6. Reverse Polish notation: `5 7 2 + 6 -`
Answer:

```
ERROR
```

Q7. Complete the following function that returns `true` if the integer parameter `n` is a palindrome. An integer is a palindrome if it's the same if the read it left-to-right or right-to-left. For instance `0`, `12321`, and `123321` are palindromes. Reading `12321` left-to-right, you see `1, 2, 3, 2, 1`, which is the same if you read it right-to-left.

But `-1` and `1232` are not palindromes. Note that reading `-1` left-to-right gives you `-, 1` and reading it right-to-left gives you `1, -`.
Answer:

```
bool is_palindrome(int n)
{
  if (n < 0) return false;

    int original = n;
    int reversed = 0;

    while (n > 0)
    {
        int digit = n % 10;
        reversed = reversed * 10 + digit;
        n /= 10;
    }

    return original == reversed;
}
```

(HINT: You can use a stack. But you don't have to. There are basically two ways to solve this problem, you either look at the digits of n, or you form the reverse of n with the digits. There are variations of the above ideas.)

Instructions

In the file `thispreamble.tex` look for

```
\renewcommand\AUTHOR{}
```

and enter your email address:

```
\renewcommand\AUTHOR{jdoe5@cougars.ccis.edu}
```

(This is not really necessary since alex will change that for you when you execute `make`.) In your bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

Enter your answers in `main.tex`. In the bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

For each question, you'll see boxes for you to fill. For small boxes, if you see

```
1 + 1 = \answerbox{}.
```

you do this:

```
1 + 1 = \answerbox{2}.
```

`answerbox` will also appear in "true/false" and "multiple-choice" questions.

For longer answers that need typewriter font, if you see

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
\end{answercode}
```

you do this:

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
int x;
\end{answercode}
```

`answercode` will appear in questions asking for code, algorithm, and program output. In this case, indentation and spacing is significant. For program output, I do look at spaces and newlines.

For long answers (not in typewriter font) if you see

```
What is the color of the sky?
\begin{answerlong}
\end{answerlong}
```

you can write

```
What is the color of the sky?
\begin{answerlong}
The color of the sky is blue.
\end{answerlong}
```

A question that begins with "T or F or M" requires you to identify whether it is true or false, or meaningless. "Meaningless" means something's wrong with the question and it is not well-defined. Something like "$1 + 2 = 4$" is either true or false (of course it's false). Something like "$1+_2 = 4$?" does not make sense.

When writing results of computations, make sure it's simplified. For instance write 2 instead of $1 + 1$.

Higher level classes.

For students beyond 245: You can put LaTeX commands in `answerlong`.

More examples of meaningless statements: Questions such as "Is $42 = 1+_2$ true or false?" or "Is $42 = \{2\}^{\{3\}}$ true or false?" does not make sense. "Is $P(42) = \{42\}$ true or false?" is meaningless because $P(X)$ is only defined if X is a set. For "Is $1 + 2 + 3$ true or false?", "$1 + 2 + 3$" is well–defined but as a "numerical expression", not as a "proposition", i.e., it cannot be true or false. Therefore "Is $1 + 2 + 3$ true or false?" is also not a well-defined question.

More examples of simplification: When you write down sets, if the answer is $\{1\}$, do not write $\{1, 1\}$. And when the values can be ordered, write the elements of the set in ascending order. When writing polynomials, begin with the highest degree term.

When writing a counterexample, always write the simplest.